Indroneel Roy

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Research Interests: *Medical Image Segmentation · Hybrid CNN-Transformer-Mamba Architectures · Vision-Language Models · Self-/Semi-Supervised Learning · Diffusion Models*

Research Summary

Computer vision researcher and EEE graduate from SUST, focused on hybrid CNN-Transformer-Mamba architectures for medical image segmentation. My work spans wavelet-guided edge attention, state-space (Mamba) sequence modelling, and dual-branch Transformer decoding for precise, efficient segmentation, with a growing interest in self-supervised and vision-language approaches to reduce reliance on dense pixel-level annotation. Seeking a PhD to take this further.

Education

Shahjalal University of Science and Technology (SUST) — February 2020 – August 2025
B.Sc. in Electrical & Electronic Engineering — ***GPA: 3.22/4.00***

Research Projects

EdgeMambaFormer: Edge-Guided Cross-Scale Mamba Transformer with Wavelet-Based Boundary Attention for Colonoscopy Polyp Segmentation | *PyTorch, Mamba (SSM), Transformers* — GitHub ↗

- Proposed a tri-hybrid CNN–Mamba–Transformer architecture built on a PVT v2-B2 encoder, combining a parameter-free Wavelet Edge Attention Gate (WaveletEAG), a bidirectional Cross-Scale Mamba Module (CSMM) for linear-complexity global context, and a dual-branch (local window + global) Transformer decoder.
- Introduced WaveletEAG, using a 2-D Haar Discrete Wavelet Transform to extract multi-directional boundary signals (horizontal, vertical, diagonal) without any learned convolutional edge detector, fused with semantic context via a learnable sigmoid gate.
- Designed CSMM, which jointly tokenises multi-scale encoder features into a single sequence and applies a bidirectional selective-scan SSM (S6) for cross-scale global dependency modelling at linear complexity.
- Achieved state-of-the-art results on 3 of 4 independently trained colonoscopy polyp benchmarks: **0.935** mDice / **0.884** mIoU on Kvasir-SEG, **0.941/0.892** on CVC-ClinicDB, and **0.889/0.812** on CVC-ColonDB (+8.1 pp mDice over Polyp-PVT), with a compact 25.4M-parameter footprint; validated each module via ablation.

MSHFNet: Multi-Scale Hybrid Fusion Network for Multi-Organ CT Segmentation | *PyTorch, timm* — GitHub ↗

- Proposed MSHFNet, a dual-pathway encoder-decoder architecture integrating a pretrained ResNet-50 CNN encoder with a Swin-Base Transformer encoder, designed for precise multi-organ segmentation on abdominal CT scans.
- Designed a novel Cross-Attention Fusion Module (CAFM) performing bidirectional cross-attention at four hierarchical scales, enabling each modality to selectively enrich the other beyond simple concatenation or addition.
- Implemented a Shared Decoder with Deep Supervision using a combined Cross-Entropy, Dice, and Boundary loss, providing auxiliary gradient signals at three intermediate resolutions to improve boundary-aware learning.
- Achieved **79.73%** mean DSC on the Synapse multi-organ benchmark (8 organs), surpassing TransUNet by +2.25% DSC, with state-of-the-art Pancreas segmentation at **81.47%**; further achieved **93.76%** mean DSC on the ACDC cardiac MRI benchmark, confirming cross-domain generalisability.

PolyFormer: Hybrid CNN-Transformer Network for Polyp Segmentation | *PyTorch, OpenCV* — GitHub ↗

- Designed a novel encoder-decoder architecture integrating a pretrained **ResNet-50 CNN** backbone with a custom **Transformer bottleneck** featuring learnable positional encodings and 8-head multi-head self-attention for global context modeling in colonoscopy images.
- Developed a **Cross-Attention Fusion Module** where Transformer features attend to CNN spatial features as key-value pairs, enabling explicit synergy between local texture representations and long-range spatial dependencies.
- Implemented a **Combined Dice + BCE loss** with a U-Net style hierarchical decoder using four skip-connection stages, improving boundary precision for small and irregularly shaped polyps.
- Achieved **Dice: 0.8971** and **IoU: 0.8172** on the **Kvasir-SEG** benchmark (60 epochs, NVIDIA T4 GPU), trained with AdamW optimizer and Cosine Annealing scheduler.

Additional Projects

Cassava Leaf Disease Classification using Vision Transformer (ViT) | *PyTorch, timm* — GitHub ↗

– Designed and fine-tuned a ViT-Base/16 model for multi-class plant disease classification across 5 categories on the Kaggle Cassava Leaf Disease dataset (21,367 training images).
– Achieved **87.15%** validation accuracy with **93.4%** precision and **95.6%** recall for Cassava Mosaic Disease (CMD), outperforming ResNet-50 baseline by ∼4.2%.
– Implemented full training pipeline with transfer learning via timm, GPU-accelerated 224×224 preprocessing, label smoothing, and cosine LR scheduling.

Vision-Based Wheat Head Detection using YOLOv11 | *PyTorch, OpenCV* GitHub ↗

– Fine-tuned YOLOv11-Medium on a large-scale wheat head detection dataset with **29,422+** annotated instances for precision agriculture applications.
– Achieved **94.7% mAP@50** and **55.4% mAP@50–95** with **91.7%** precision and **89.2%** recall, demonstrating strong generalization under varying occlusion and lighting.

Agentflow: Multi-Tool Conversational AI System | *LangChain, LangGraph, LLM* GitHub ↗

– Designed a stateful conversational AI system using LangGraph state machine architecture with persistent SQLite checkpointing for robust multi-turn dialogue management.
– Integrated 10+ specialized tools including web search, financial APIs (Alpha Vantage, CoinGecko), and weather services with dynamic tool binding using Groq LLM (Llama-3.3-70B).

Research & Model Implementations

- **CNN Architectures** — Implemented and trained LeNet, AlexNet, ResNet, DenseNet, and EfficientNet from scratch in PyTorch; performed architectural comparisons and benchmarked on standard image classification tasks.
- **Transformer-Based Models** — Implemented Transformer, Swin Transformer, and TransUNet; investigated self-attention mechanisms, window-based attention, and hierarchical vision representations for classification and segmentation.
- **State-Space / Mamba Models** — Implemented bidirectional selective-scan (S6) Mamba blocks for cross-scale vision sequence modelling, investigating linear-complexity alternatives to self-attention for dense prediction tasks.
- **Image Segmentation Models** — Implemented U-Net, SegFormer, and MaskFormer; conducted experiments on medical and natural image segmentation datasets using Dice and IoU metrics.
- **Diffusion Models** — Implemented DDPM, DDIM, Classifier-Free Guidance, and Latent Diffusion Model (LDM) from scratch, covering the noise-schedule, accelerated sampling, and latent-space formulations underlying modern generative diffusion; further explored LoRA-adapted Stable Diffusion U-Net backbones for medical image segmentation. GitHub ↗.
- **Code Repository** — All implementations publicly available and reproducible on GitHub ↗.

Technical Skills

Languages: Python, SQL, C, MATLAB, Assembly (Intel ×86)

ML/DL Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV

Computer Vision: CNNs, Object Detection (YOLO), Image Segmentation, ResNet, U-Net, ViT, Swin Transformer, Mamba/SSM

LLM & NLP: LangChain, LangGraph, Transformers, Hugging Face, RAG, Vector Databases

MLOps & Deployment: Docker, FastAPI, Streamlit, Hugging Face Spaces, Git, AWS

Databases: SQL, Pinecone, ChromaDB

Algorithms & DS: Data Structures & Algorithms 200+ LeetCode problems solved

Certifications

Neural Networks and Deep Learning — Coursera / deeplearning.ai (View Certificate)

Additional Information

Languages: English (Fluent) · Bengali (Native)
Interests: Open-source AI contributions · Research paper implementations · Technical blogging · Chess · Cooking